

XRPL Validator — Daily Engineering Report • 2026-09-06

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **12 commits** (xrpl-validator 12)
- **12 engine findings fixed** (F185–F197)
- Amendment **fixCleanup3_3_0** held majority (since 2026-08-28 11:15 UTC); activation due ~2026-09-11 11:15 UTC

Engine findings fixed

- **fix(xrpl-ledger): F185 — an NFTokenCreateOffer with a past Expiration is tecEXPIRED** (a8e8cde, xrpl-validator) `hasExpired(ctx.view, ctx.tx[~sfExpiration])` — the parent ledger's close time at or past the offer's Expiration — is the FIRST test of `NFTokenCreateOffer::preclaim` (`NFTokenCreateOffer.cpp`), before the token is looked up. The native transactor had no expiration test and created the offer.
- **fix(xrpl-ledger): F187 — a created DID object carries Flags: 0** (0b4ea2a, xrpl-validator) `sfFlags` is `soeREQUIRED` on the DID ledger format, so rippled serialises `Flags: 0 (22 00000000)` right after the entry type of every DID it creates, whether or not `DIDSet` touches it. Our `DIDSet` built the object without the field.
- **fix(xrpl-ledger): F186 — an in-limited multi-path AMM slice truncates its out** (347380f, xrpl-validator) `limitStepIn` hands the in-limited slice to `AMMOffer::limitIn(..., roundUp=false)`, which in multi-path mode is `Quality::ceilInStrict` → `divRoundStrict(limit, rate, false)`: the 17-digit integer quotient TRUNCATED to 16 digits (`STAmount.cpp` `divRoundImpl`, `roundUp=false` — no bump in `muldivRound`, canonicalizeRound down). Our slice sizing floored a ~17-digit quotient in `me_muldiv` and re-normalised to 16 digits at ...
- **fix(xrpl-ledger): F188 — TrustSet quality fields: side-specific, absent when zero** (2502ad8, xrpl-validator) `TrustSet::doApply` writes the sender-side quality — `bHigh ? sfHighQualityIn : sfLowQualityIn` (and Out) — only when the transaction's value is non-zero and not `QUALITY_ONE`; zero (or 1e9) `makeFieldAbsent`s the field. Ours copied the transaction's `QualityIn/QualityOut` onto the line JSON under the `TRANSACTION` field names, so a `RippleState` went out carrying `sfQualityIn` (20) and `sfQualityOut` (21) — fields that do ...
- **fix(xrpl-ledger): F189 — AccountDelete's age rule is Sequence + 255 against the ledger being built** (d923dee, xrpl-validator) `(*sleAccount)[sfSequence] + kSeqDelta > ctx.view.seq()` → `tecTOO_SOON`, with `kSeqDelta = 255` and `view.seq()` the ledger under construction — the parent's sequence plus one (`AccountDelete.cpp`:287-289). Ours compared `Sequence + 256` against the PARENT and answered `tecNO_PERMISSION`: two ledgers too strict, and the wrong code. `TxResult::TooSoon` ("tecTOO_SOON", fee-claiming) is added for it.
- **fix(xrpl-ledger): F190 — an implied inter-gateway tail keeps a path's strand active** (d9c5a54, xrpl-validator) `toStrand`'s normalization appends the deliver issuer as an `ACCOUNT` element when the path ends

holding the delivered currency under another issuer (PaySteps.cpp), so the strand closes with DirectSteps across the two gateways' mutual line — a real step, checked by DirectStepI::check (no line $\Rightarrow$ the strand is not built). Our `mixed_layout` counted only the path's own account elements as "real" runs, so this implied tail ...

- **fix(xrpl-ledger): F191 — self-offers beyond the strict limit: the pool pre-empts the tip only from inside the limit** (`fcfb7fc` , `xrpl-validator`) Two rules of the beyond-strict level walk in offer crossing, both from BookStep.cpp's stream loop: * The pool's turn (`tryAMM(offers.tip().quality())`) pre-empts the level's CLOB offers only when its anchored offer is itself INSIDE the limit. Beyond it the offer fails the quality threshold, nothing is consumed, `execOffer` steps onto the tip, and a self-offer there is removed (`limitSelfCrossQuality` , "Remove this ...
- **fix(xrpl-ledger): F192+F193 — XRP received in a flow does not fund the same flow** (`6617922` , `xrpl-validator`) Two findings from the #106804064–#106804073 receipts, both in offer.rs.
- **fix(xrpl-ledger): F194 — a fee-less exhausting slice is debited the folded remainder** (`f562fff` , `xrpl-validator`) rippled's in-limited fill debits the taker `stpAmt.in = remainingIn`, and `remainingIn` is `SendMax` less the ascending 16-digit fold of the saved per-iteration ins (StrandFlow.h). The walk's `rem_gets` already re-derives that fold at every iteration boundary, but the AMM turn's remaining GROSS cap — the figure an exhausting slice is debited verbatim (the in-limited gross-primary rule, #106455039) — came from the ...
- **fix(xrpl-ledger): F196 — CredentialCreate refuses a past Expiration; a mint's offer half expires like NFTokenCreateOffer's** (`7fb9c54` , `xrpl-validator`) Two rippled 3.3.0 checks ported line-for-line, no mainnet specimen yet (both were listed as open no-specimen gaps after F182/F185): * `CredentialCreate::doApply` — `closeTime > *optExp` $\rightarrow$ `tecEXPIRED`, judged before the reserve check. STRICT: an Expiration equal to the parent close still creates. No `exp != 0` guard. * `NFTokenMint::preclaim` — `hasExpired(ctx.view, ctx.tx[~sfExpiration])` is the first test of the Amount ...
- **fix(xrpl-ledger): F197 — a bridge walk's exhausting fill is bounded by the taker's line as the ledger carries it** (`dcf0734` , `xrpl-validator`) rippled's DirectStep limits an iteration's in by `maxSrcToDst` , read through `PaymentSandbox::balanceHookIOU = min(live line balance, original -  $\Sigma$ debits)`. The exhausting slice's verbatim gross remainder (F50/F51: budget – the 16-digit fold of the spends) drifts an ulp from the sequential per-iteration line chain on a long crossing, and rippled takes the smaller. The direct walk had this bound (F122/F133/F169b); the ...

Docs and operations

- **chore: remove the Glow submission draft from the public tree** (`6d31efc` , `xrpl-validator`)

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Size-Weighted Owner Reserve Units** (fees) Add a per-LedgerEntryType weight table (a protocol constant: 1 unit for fixed-size objects like Offer/Escrow, ceil(serializedSize/192 bytes) for varia
- **MPT Issuer Holder-Cap Enforcement** (rwa) Add sfMaxHolderCount and sfHolderCount SFields to MPTokenIssuance plus a tfHolderCapped flag marking the issuance as cap-enforced. MPTokenAuthorize's
- **Policy-Scoped Pseudo-Account Session Keys** (accounts) Add a SessionKeySet transaction that creates an owner-reserved ltSESSION_KEY object (keylet::sessionKey(account, seq)) carrying sfSessionPublicKey, sf

Appendix — commits

Time	Repo	Commit	Subject
00:38	xrpl-validator	a8e8cde	fix(xrpl-ledger): F185 — an NFTokenCreateOffer with a past Expiration is tecEXPIRED
04:40	xrpl-validator	0b4ea2a	fix(xrpl-ledger): F187 — a created DID object carries Flags: 0
04:40	xrpl-validator	347380f	fix(xrpl-ledger): F186 — an in-limited multi-path AMM slice truncates its out
08:43	xrpl-validator	2502ad8	fix(xrpl-ledger): F188 — TrustSet quality fields: side-specific, absent when zero
09:03	xrpl-validator	d923dee	fix(xrpl-ledger): F189 — AccountDelete's age rule is Sequence + 255 against the ledger being built
09:45	xrpl-validator	d9c5a54	fix(xrpl-ledger): F190 — an implied inter-gateway tail keeps a path's strand active
11:44	xrpl-validator	fcfb7fc	fix(xrpl-ledger): F191 — self-offers beyond the strict limit: the pool pre-empts the tip only from inside the limit
12:49	xrpl-validator	6617922	fix(xrpl-ledger): F192+F193 — XRP received in a flow does not fund the same flow
12:57	xrpl-validator	f562fff	fix(xrpl-ledger): F194 — a fee-less exhausting slice is debited the folded remainder
14:24	xrpl-validator	7fb9c54	fix(xrpl-ledger): F196 — CredentialCreate refuses a past Expiration; a mint's offer half expires like NFTokenCreateOffer's
23:36	xrpl-validator	dcf0734	fix(xrpl-ledger): F197 — a bridge walk's exhausting fill is bounded by the taker's line as the ledger carries it
23:40	xrpl-validator	6d31efc	chore: remove the Glow submission draft from the public tree